

Model Predictive Control

Chapter 10: Robustness of Nominal MPC

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Outline

1. Robustness of Nominal MPC

Nominal MPC with Noise

We want to control the noisy system:

$$x(k+1) = Ax(k) + Bu(k) + w(k)$$

What happens if we just ignore the noise and hope for the best?

Setup and solve a standard MPC problem:

$$\begin{aligned} J^*(x_0) &= \min_U \sum_{i=0}^{N-1} l(x_i, u_i) + l_f(x_N) \\ \text{subj. to } &x_{i+1} = Ax_i + Bu_i \\ &x_i, u_i \in \mathcal{X} \times \mathcal{U} \\ &x_N \in \mathcal{X}_f \end{aligned}$$

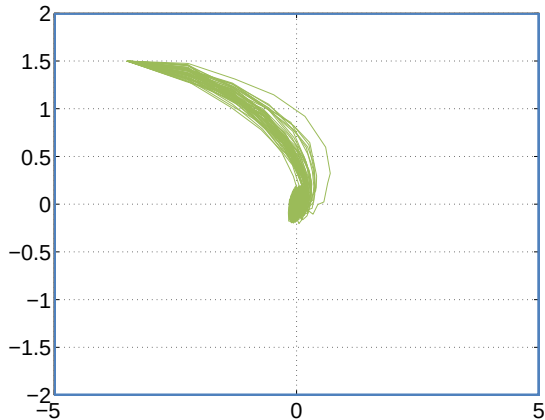
Our closed-loop system is now:

$$x(k+1) = Ax(k) + Bu_0^*(x(k)) + w(k)$$

⇒ Can prove convergence to a neighborhood of the origin (for linear systems s.t. polytopic constraints!)

Example

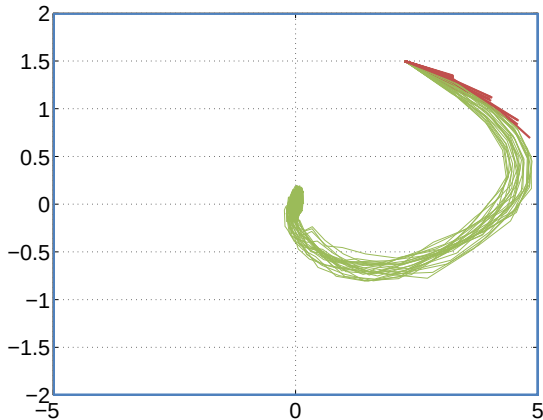
Consider system with noise, but we pretend it's not there in the controller.



- 100 trajectories with different noise realizations
- Seems to work fine?!

Example

Consider system with noise, but we pretend it's not there in the controller.



- 100 trajectories with different noise realizations
- Seems to work fine?!
- Can no longer be certain it will work!
- For some states it will work sometimes

How do we formalize this idea?

What Happens to Our Lyapunov Function?

Recall: The optimal cost $J^*(x)$ is a Lyapunov function for the nominal system

$$J^*(Ax + Bu^*(x)) - J^*(x) \leq -l(x, u^*(x))$$

However, our state at the next point in time is now

$$x(k+1) = Ax(k) + Bu^*(x(k)) + w(k)$$

Do we still have a Lyapunov decrease?

What Happens to Our Lyapunov Function?

Assume: Optimal cost J^* is Lipschitz continuous

$$\begin{aligned} |J^*(Ax + Bu^*(x) + w) - J^*(Ax + Bu^*(x))| \\ \leq \gamma \|Ax + Bu^*(x) + w - (Ax + Bu^*(x))\| = \gamma \|w\| \end{aligned}$$

Our Lyapunov decrease can be bounded as:

$$\begin{aligned} J^*(Ax + Bu^*(x) + w) - J^*(x) \\ = J^*(Ax + Bu^*(x) + w) - J^*(x) - J^*(Ax + Bu^*(x)) + J^*(Ax + Bu^*(x)) \\ \leq J^*(Ax + Bu^*(x)) - J^*(x) + \gamma \|w\| \\ \leq -l(x, u^*(x)) + \gamma \|w\| \end{aligned}$$

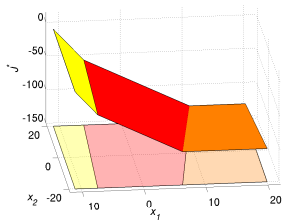
- Amount of decrease grows with $\|x\|$
- Amount of increase is upper bounded by $\max \{\|w\| \mid w \in \mathcal{W}\}$

Therefore we will move towards the origin until there is a balance between the size of x and the size of w

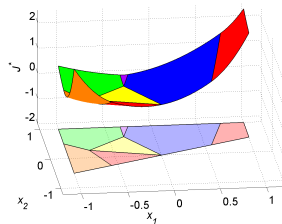
Special case when nominal MPC is robust

If the nominal model is linear, the cost function $V_N(x, U)$ is linear or quadratic, and the sets of constraints are polytopes, then the optimal cost $V_N^*(x)$ is Lipschitz continuous in $\mathcal{X}_N$.

Recall: In this case the optimal cost function is a piece-wise linear/quadratic function¹. Then $V_N^*(x)$ is Lipschitz in $\mathcal{X}_N$.



Example optimal cost for linear cost function



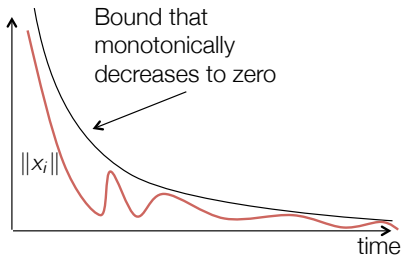
Example optimal cost for quadratic cost function

¹Bemporad et al., "Model predictive control based on linear programming – the explicit solution". IEEE Trans. on Automatic Control, 47(12), 1974–1985, 2002.

Input-to-State Stability

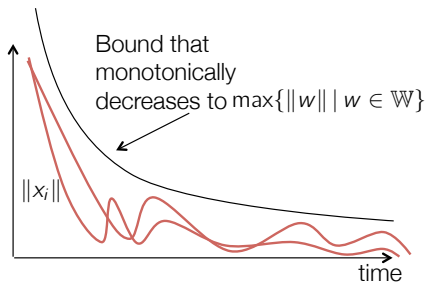
What we have shown is that our system is **Input-to-State Stable**.²

Asymptotic stability



System converges to zero

ISS stability



Converges to set around zero, whose size is determined by size of the noise

² Limon, D., Alamo, T., Raimondo, D. M., Muñoz de la Peña, D., Bravo, J. M., Ferramosca, A., and Camacho, E. F. (2009). Input-to-State Stability: A Unifying Framework for Robust Model Predictive Control. In L. Magni, D. M. Raimondo, & F. Allgöwer (Eds.), Nonlinear Model Predictive Control (Vol. 384, pp. 1-26). Berlin, Heidelberg: Springer Berlin Heidelberg. doi:10.1007/978-3-642-01094-1

Nominal MPC for Uncertain Systems - Summary

Idea

- Ignore the noise and hope it works

Benefits

- Simple
- No knowledge of the noise set $\mathcal{W}$ required - 'just works'
- Often very effective in practice (this is what most practitioners do anyway)
- Feasible set is large (we can find a solution, but it may not work)
- Region of attraction may be larger than other approaches

Cons

- Very difficult to determine region of attraction (set of states in which the controller works)
- Hard to tune - no obvious way to tradeoff robustness against performance
- Works for linear systems, for nonlinear systems only under continuity assumptions